Radboud Universiteit

Bayesian regression for language scientists

Lecture 2

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LOT2026 Summer School

Outline

- Basic principles: Q&A
- Linear regression:
  - ordinary least squares model with lm()
  - Bayesian model with brm()
- Simulating data for mixed models
- Linear mixed regression:
  - (restricted) maximum likelihood model with lmer()
  - Bayesian model with brm()

Bayesian statistics and statistical power

- If there is a true effect of size X, what is the probability that my test will reject the null hypothesis?
- Type II errors in frequentist statistics
- No direct equivalent in Bayesian statistics
- Diagnostics:
  - how spread out are the posterior distibutions?
  - do the priors influence the posteriors too much? (prior sensitivity analysis)
  - do the coefficients remain stable?
- Bayesian statistics is not magic.

Bayesian statistics and the replicability crisis

- Different strategies of improving the situation:
  - Replication (Bayesian or not)
  - Better design, larger datasets, more transparency...
  - Bayesian cumulative science:
    - using informative priors helps to avoid "false alarm" (5% Type I error)
    - knowledge is not "reset" every time

How to choose priors?

- Non-informative priors: not recommended, unless you want to replicate a frequentist analysis as faithfully as possible.
- Weakly informative: the most popular, nearly always justified (AKA regularizing priors)
- Strong (Assertia's) priors: only based on a very similar, well-understood model (example on Wednesday)
- It is always recommended to perform a prior sensitivity analysis.

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Imaginary data

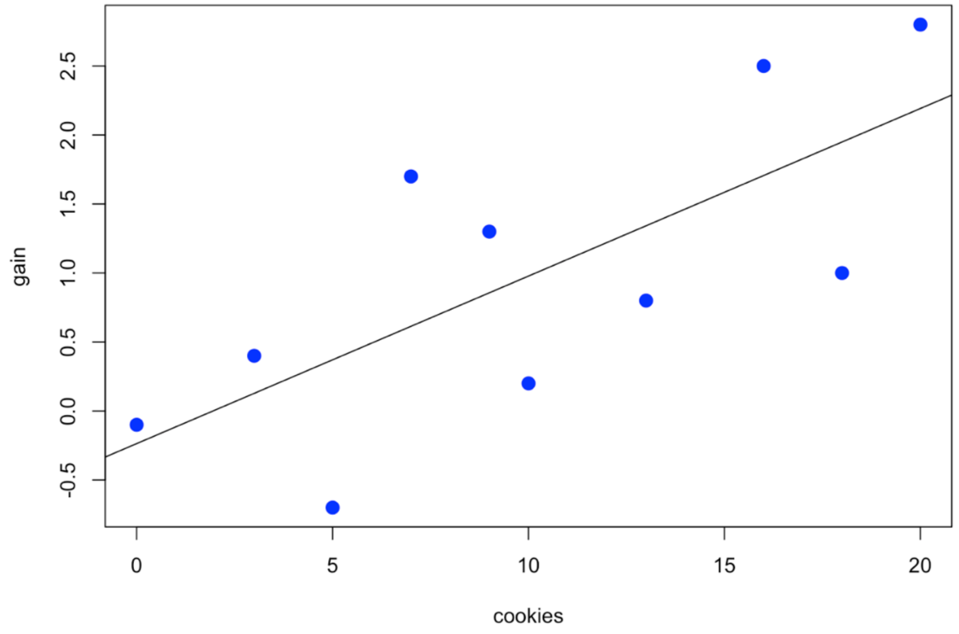
Name	Cookies eaten per day	Kilos gained
John	0	-0.1
Mary	3	0.4
Bill	5	-0.7
Jane	7	1.7
Laura	9	1.3
Ann	10	0.2
Chris	13	0.8
Eve	16	2.5
Peter	18	1.0
Steve	20	2.8

Cookies data in R

```
cookies <- c(0, 3, 5, 7, 9, 10, 13, 16, 18, 20)
gain <- c(-0.1, 0.4, -0.7,  1.7, 1.3, 0.2,
          0.8, 2.5, 1.0, 2.8)
df_cookies <- data.frame(gain, cookies)
```

Plotting the data

```
plot(cookies, gain, pch = 16, col = "blue", cex = 1.5)
abline(lm(gain ~ cookies))
```



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A traditional linear model (OLS)

```
m_ols <- lm(gain ~ cookies, data = df_cookies)
summary(m_ols)
```

```
…
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.23645    0.49341  -0.479   0.6446
cookies      0.12143    0.04151   2.925   0.0191 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

TOTAL RECALL

What are betas in regression?

Frequentist statistics	Bayesian statistics
Parameters as unknown but fixed constants A coefficient estimate in regression is one number.	Parameters as random variables represented as probability distributions A coefficient estimate is a probability distribution.

95% confidence intervals

```
confint(m_ols)
                  2.5 %    97.5 %
(Intercept) -1.37424891 0.9013550
cookies      0.02571195 0.2171488
```

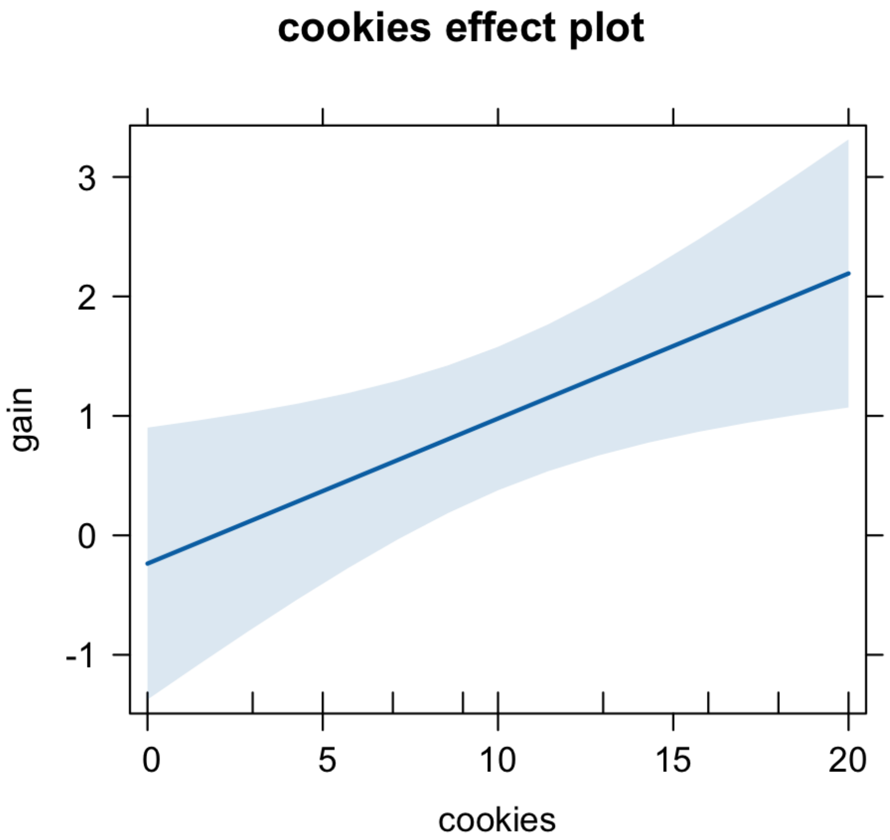
TOTAL RECALL

Intervals

Frequentist statistics	Bayesian statistics
Confidence intervals: If we calculate a 95% confidence interval in this way many times on repeated samples, the true mean will be between the boundaries 95% of the time.	**Credible intervals:** the parameter of interest (e.g. a regression coefficient) falls in this interval with the probability of … % Different types: a) Highest Posterior Density b) Equal-Tailed Interval

Visualization of a slope

```
library(effects)
plot(effect("cookies", m_ols))
```



Model selection

- Does doing sports affect weight gain?
- Let's add another variable:

```
df_cookies$sports <- c("Yes", "No", "Yes", "No", "No",
"Yes", "No", "No", "Yes", "No")
m_ols1 <- update(m_ols, . ~ . + sports)
```

F-test for model comparison

```
anova(m_ols, m_ols1)
Analysis of Variance Table

Model 1: gain ~ cookies
Model 2: gain ~ cookies + sports
  Res.Df    RSS Df Sum of Sq    F  Pr(>F)
1      8 5.4156
2      7 2.2823  1    3.1333 9.61 0.01732 *
```

AIC

```
AIC(m_ols)
[1] 28.24568

AIC(m_ols1)
[1] 21.60474
```

Linear regression diagnostics quiz

1. Are the residuals normally distributed?
2. Linearity between predictors and response
3. Homoscedasticity (equal variance along ranges of predictors)
4. Outliers
5. Influential observations
6. Multicollinearity

A. Cook's distances
B. Standardized residuals
C. QQ-plot
D. Predictor against residuals plot
E. VIF-factors
F. Component-vs-residuals plot

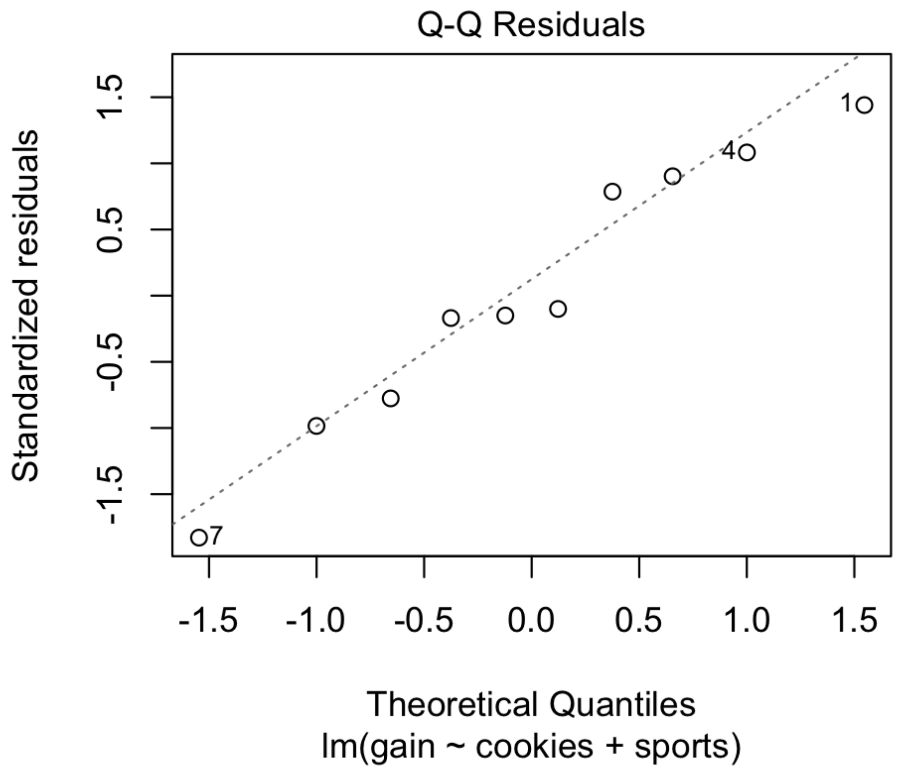
See *How to Do Linguistics with R*, Ch. 7

Some simple diagnostics

```
plot(m_ols1)
```

[many useful diagnostic plots]

```
library(car)
vif(m_ols)
 cookies   sports
1.061653 1.061653
```



Goodness of fit

```
summary(m_ols1)
...

Multiple R-squared:  0.7964,    Adjusted R-squared:  0.7382

…

fitted(m_ols1)[1:10]
 1          2          3          4          5          6
-0.7188696  0.7561923 -0.2225850  1.1532200  1.3517338  0.2736996
         7          8          9         10
 1.7487615  2.0465323  1.0677550  2.4435600
```

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MCMC parameters in brms

- number of chains *chains*. By default, 4.
- number of samples *iter*. By default, 2000.
- warm-up period *warmup* when the sampling is adjusted. The results are not used. By default, 50% of *iter*.
- Samples=chains$*$(iter−warmup)

A model with 1 chain and 20 iterations

```
library(brms)
m_brm <- brm(gain ~ cookies, data = df_cookies, chains =
1, iter = 20)
Compiling Stan program...
Start sampling
SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 1.7e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per
transition would take 0.17 seconds.
Chain 1: Adjust your expectations accordingly!
```

```
...
Warning messages:
1: There were 1 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.
2: There were 1 chains where the estimated Bayesian Fraction of Missing Information was low. See
https://mc-stan.org/misc/warnings.html#bfmi-low
3: Examine the pairs() plot to diagnose sampling problems

4: The largest R-hat is 1.9, indicating chains have not mixed.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#r-hat
5: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians may be unreliable.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess
6: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tail quantiles may be
unreliable.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

Components of summary.brmsfit (1)

```
summary(m_brm)
```

```
Family: gaussian
  Links: mu = identity
Formula: gain ~ cookies
   Data: df_cookies (Number of observations: 10)
  Draws: 1 chains, each with iter = 20; warmup = 10; thin = 1;
         total post-warmup draws = 10
```

Components of summary.brmsfit (2)

```
summary(m_brm)
…
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -1.13      1.23    -3.16     0.26 1.09        5        5
cookies       0.15      0.10    -0.00     0.30 1.04        5        5

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     1.59      0.99     0.92     3.51 1.93        5        5
```

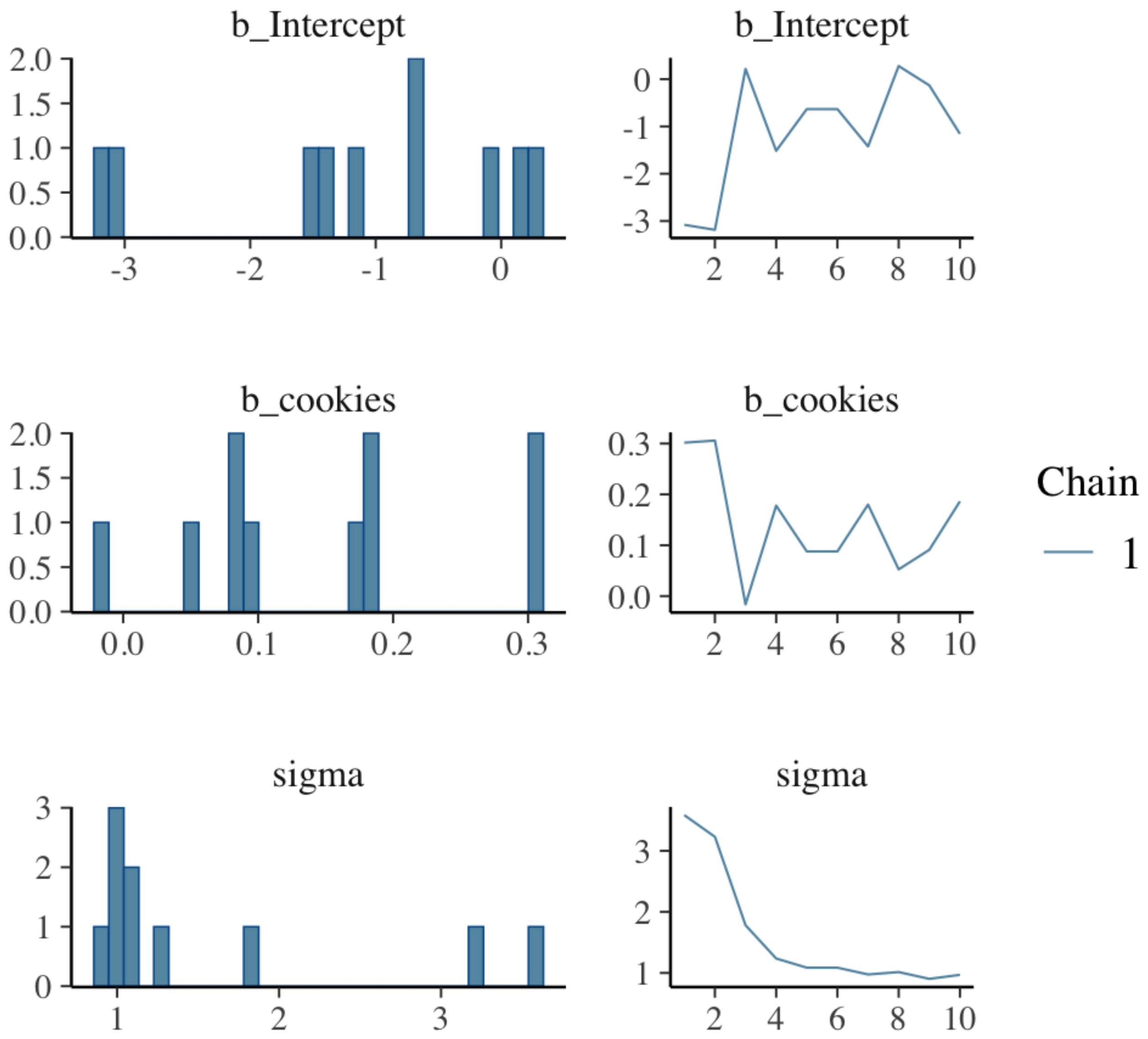
Components of summary.brmsfit (3)

…

```
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
Warning messages:
1: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations
and/or setting stronger priors.
2: There were 1 divergent transitions after warmup. Increasing adapt_delta
above 0.8 may help. See http://mc-stan.org/misc/warnings.html#divergent-
transitions-after-warmup
```

Plotting the chain

```
plot(m_brm)
```



Posterior samples

```
as_draws_df(m_brm)$b_Intercept
```

```
 [1] -3.0807885 -3.1862846  0.2153828 -1.5160944 -0.6338566 -0.6338566
 [7] -1.4232374  0.2770989 -0.1319681 -1.1575541
```

```
as_draws_df(m_brm)$b_cookies
```

```
[1]  0.30173647  0.30585607 -0.01618513  0.17782215  0.08783542
[6]  0.08783542  0.17990200  0.05258380  0.09105389  0.18629470
```

```
as_draws_df(m_brm)$sigma
```

```
[1] 3.5857493 3.2296589 1.7816557 1.2362002 1.0861594 1.0861594 0.9749818
 [8] 1.0139491 0.9026539 0.9687397
```

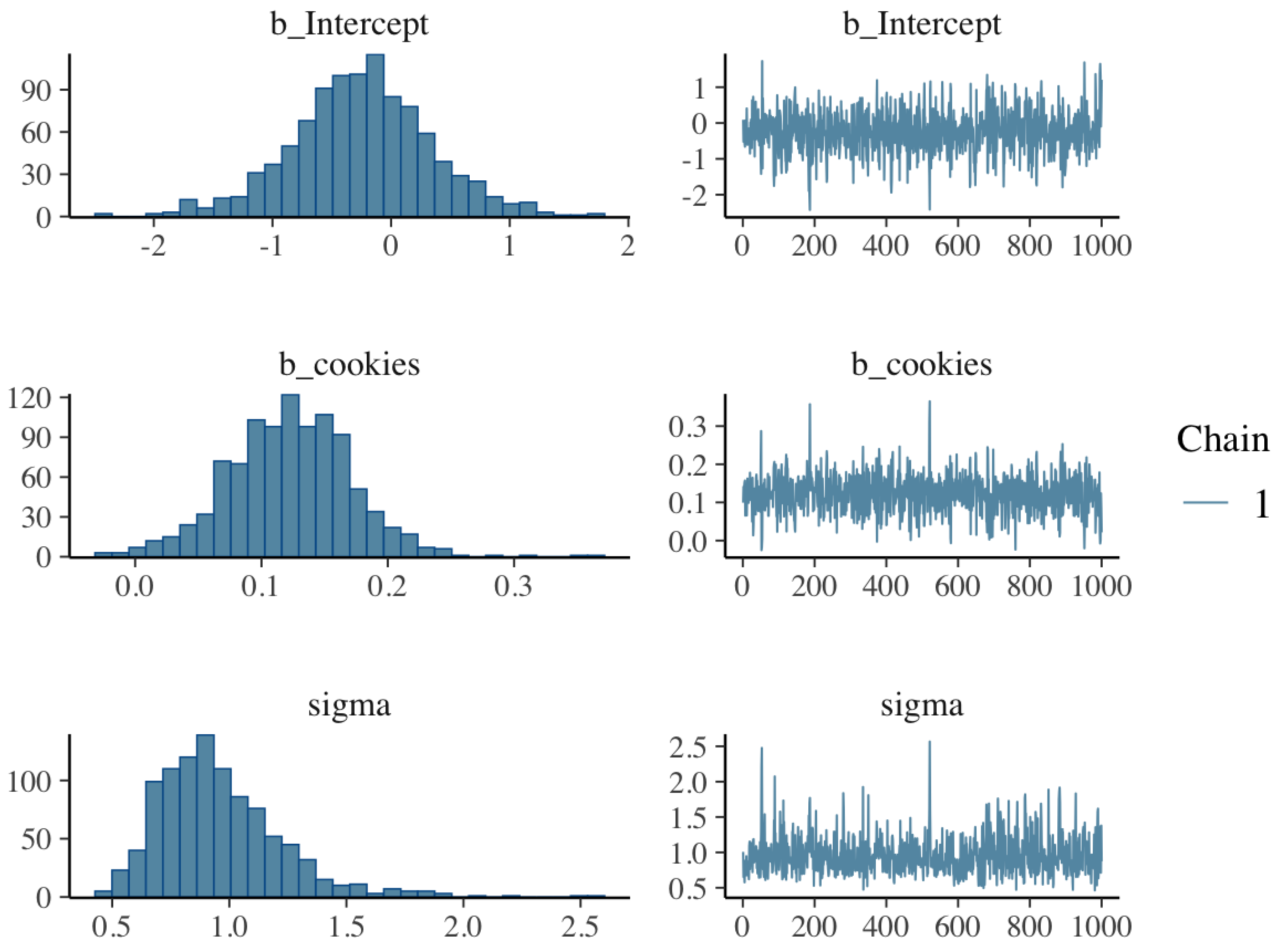
Criteria of a healthy Markov Chain

- The chain should look like a caterpillar, which does not bend. The mean should remain stable (in other words, it has reached the stationary distribution).
- the ‘caterpillar’ is hairy all over: it traverses the space quickly, without getting stuck anywhere.

More iterations (samples)

```
m_brm1 <- brm(gain ~
cookies, data = df_cookies,
chains = 1, iter = 2000)

plot(m_brm1)
```



Useful MCMC diagnostic statistics

- **R-hat**: measure of convergence of the Markov chains to the target distribution. Should be 1.00.
- **Effective sample size** in the bulk and the tail of the posterior bulk_ESS and tail_ESS: the estimate of independent samples from the posterior distribution, controlled for autocorrelation. Should be at least a few hundreds and not much lower than the number of actual samples.
- Note: R-hat is reliable for making decisions about the quality of the chain only if each of the chains has an average ESS estimate of at least 50.

Is the new model better? Why (not)?

```
summary(m_brm1)
…
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -0.25      0.59    -1.47     0.94 1.00      736      598
cookies       0.12      0.05     0.02     0.22 1.00      774      676
Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.96      0.27     0.56     1.62 1.01      592      539
```

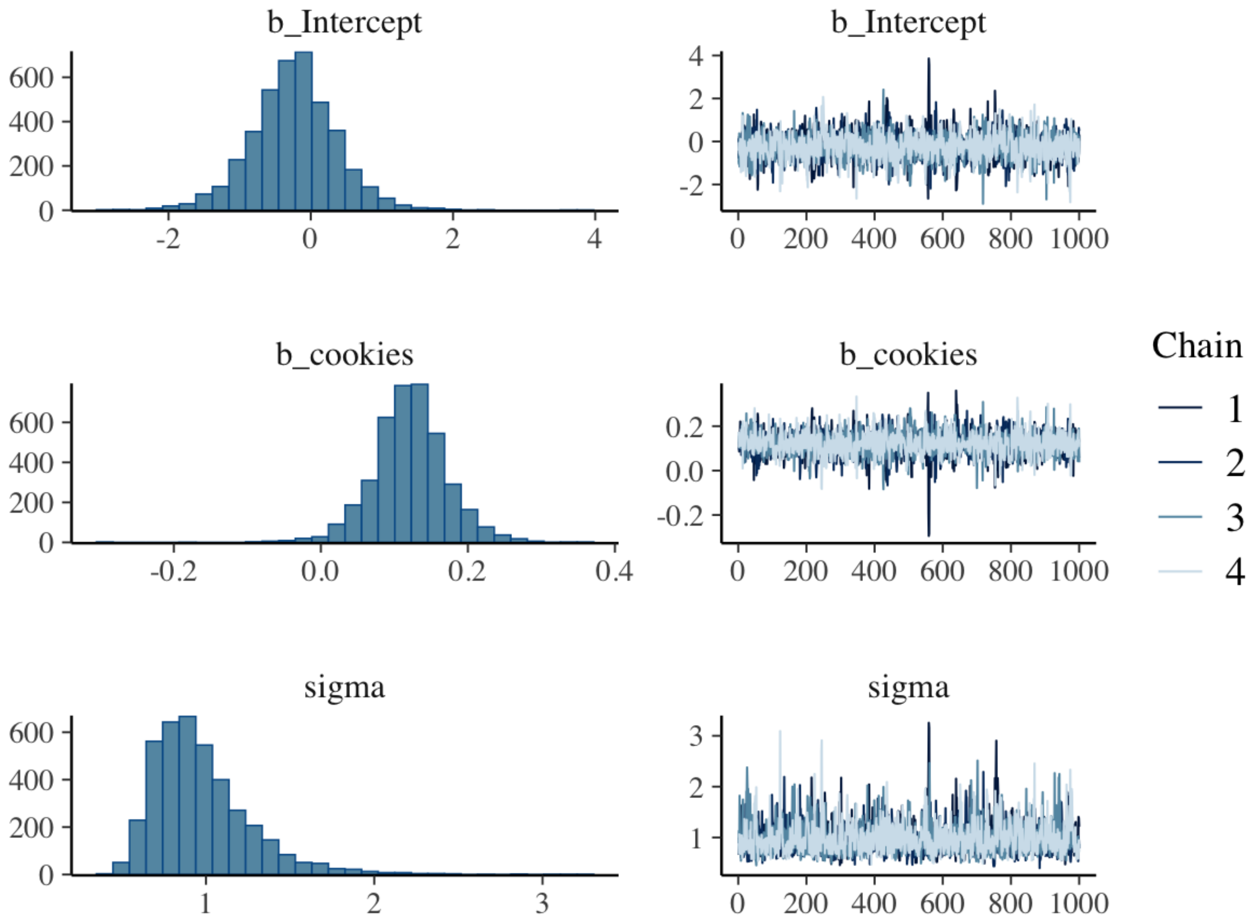
Adding more chains

- A useful tip: To save time, you can add `brm(… cores = 4)` to execute the chains in parallel.

```
m_brm2 <- brm(gain ~ cookies, data = df_cookies, chains
= 4, iter = 2000)
```

```
plot(m_brm2)
```

What has changed?



Check the R-hats and ESS values. What has changed?

MCMC rules of thumb

- There’s a misconception that Bayesian models have no convergence issues. They do!

- In the beginning of your data analysis, use multiple and shorter chains for diagnostics of any problems with your model. 4 chains (the default) are sufficient.

- Use longer chains (or even one chain) for fine-tuning the estimates you want to report.

Summarizing the posteriors: the slope

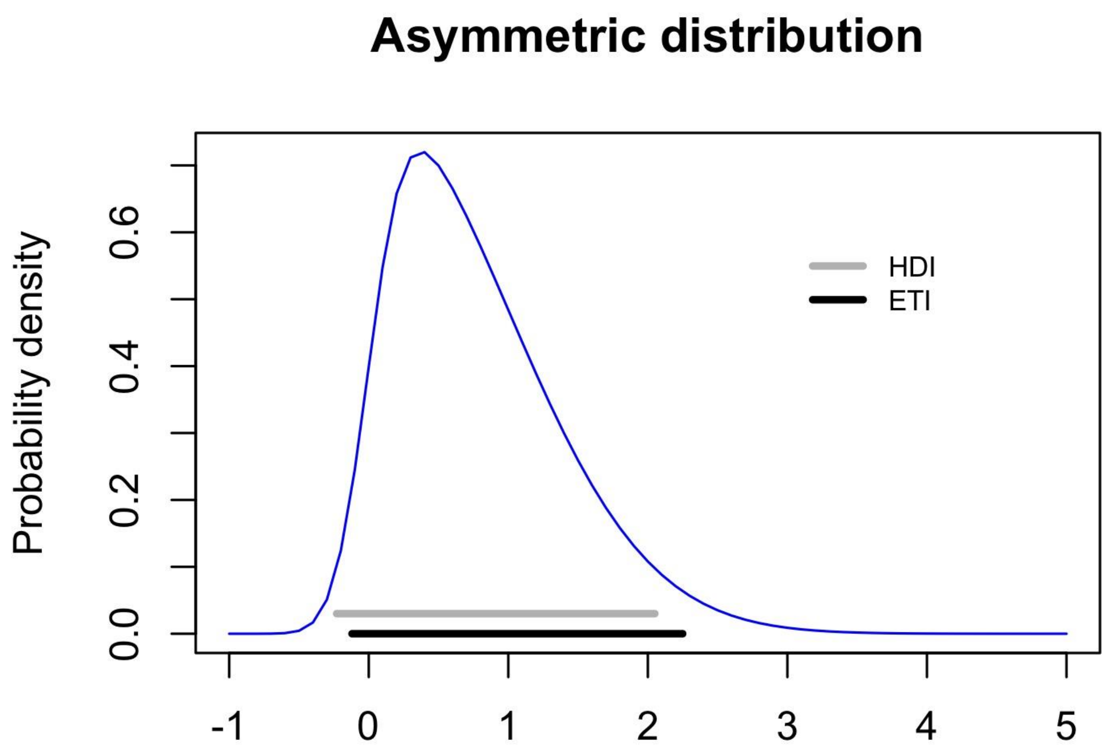
```
posterior_cookies <- as_draws_df(m_brm2)$b_cookies
mean(posterior_cookies)
[1] 0.1211222
median(posterior_cookies)
[1] 0.1211818
sd(posterior_cookies)
[1] 0.05131309
```

Compare with information in the summary! Where can you find some of these numbers?

Intervals

Frequentist statistics	Bayesian statistics
Confidence intervals: If we calculate a 95% confidence interval in this way many times on repeated samples, the true mean will be between the boundaries 95% of the time.	**Credible intervals:** the parameter of interest (e.g. a regression coefficient) falls in this interval with the probability of … % Different types: a) Highest Posterior Density b) Equal-Tailed Interval

HDI vs. ETI



Asymmetric distribution
Probability density
0.0
0.2
0.4
0.6
-1
0
1
2
3
4
5
HDI
ETI

95% Credible Intervals

- Equal-tailed Interval

```
quantile(posterior_cookies, c(0.025, 0.975))
     2.5%      97.5%
0.02157059 0.22434024
```

- Highest Density Interval

```
library(bayestestR) #not available in brms ((
hdi(posterior_cookies, 0.95)
95% HDI: [0.02, 0.22]
```

Other credible intervals

- Some statisticians believe that it is a good idea to use cut-off points different from 0.95, to prevent unconsciosly conducting null hypothesis significant tests (e.g., McElreath 2018: 58).
- For example, you can use 0.69, 0.89 and 0.97.

TOTAL RECALL

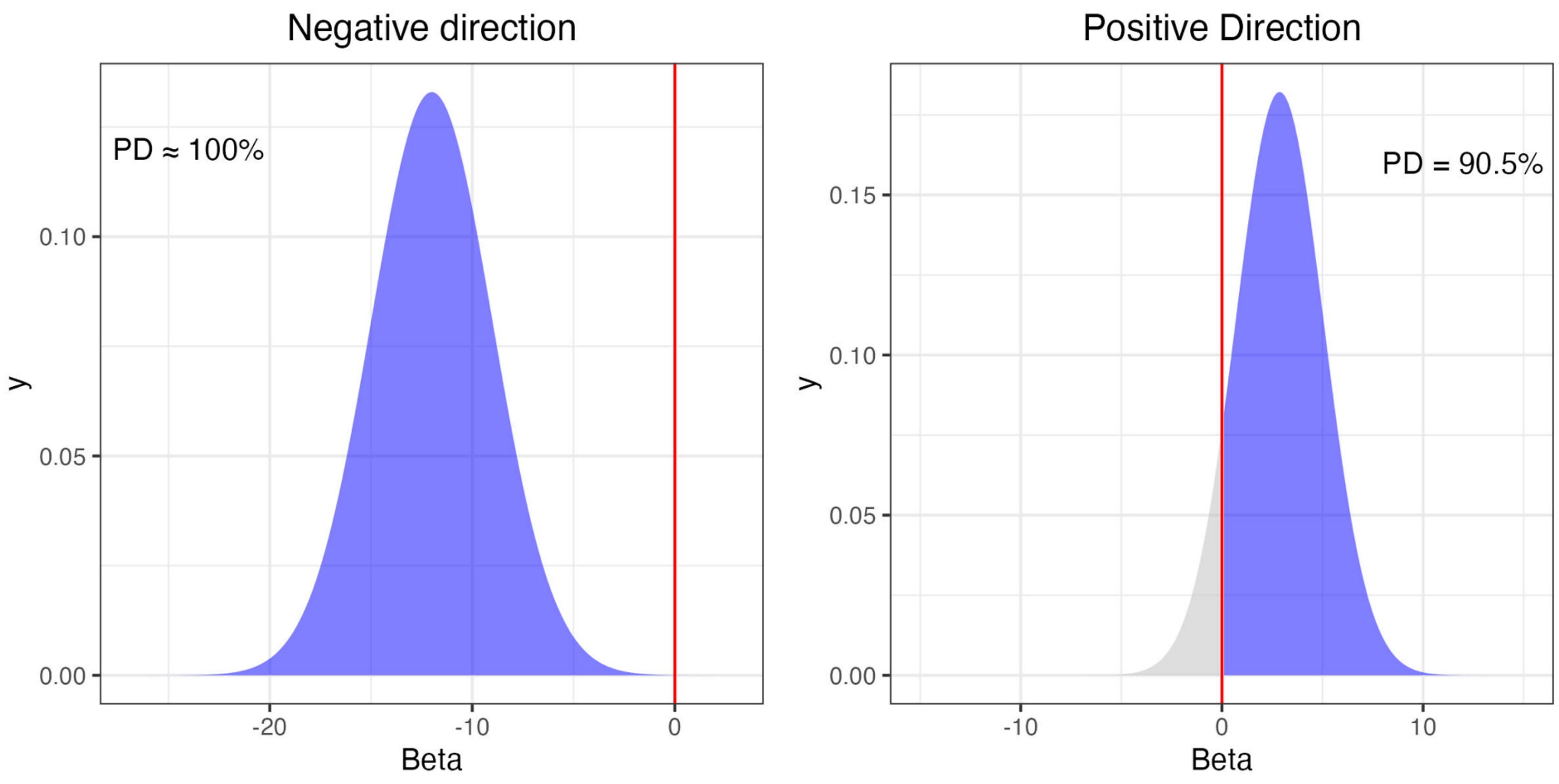
Hypothesis testing

Frequentist statistics	Bayesian statistics
Null hypothesis significance testing: Can we reject the null hypothesis? Tools: p-values, confidence intervals.	Testing the research hypothesis directly: Can we support the research hypothesis? Tools: ROPE, Probability of Direction Which hypothesis is more supported by the data? Tools: Bayes Factor, ROPE...

Is there an effect?

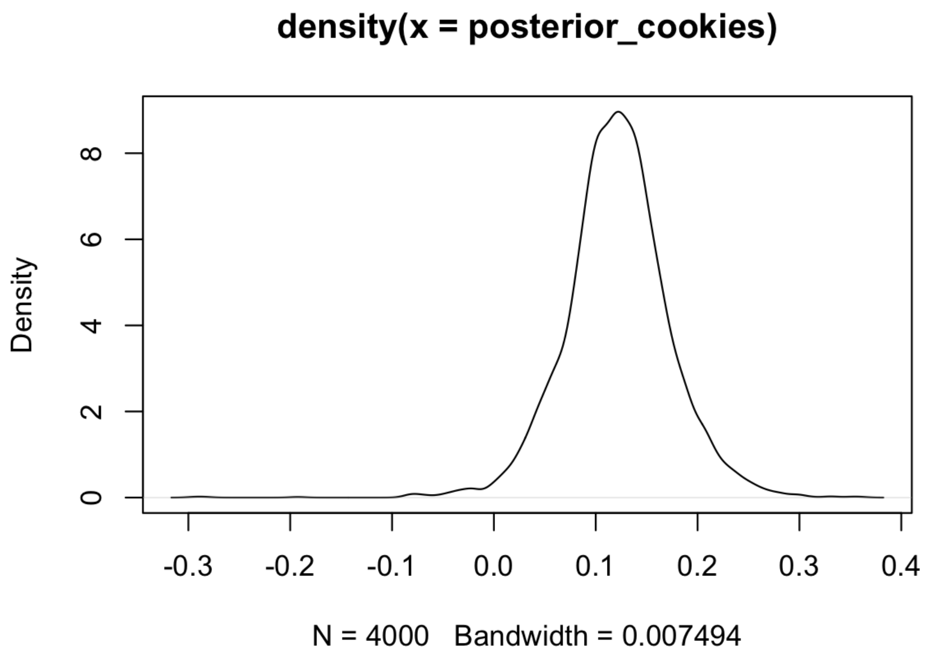
- There are many methods for making a decision if an effect is supported by the data or not: Bayes Factor, MAP-based p-value, Probability of Direction (PD)…
- The simplest measure, which also does not dependent on how you defined your priors and other parameters, is Probability of Direction. It shows the proportion of posterior distribution that has a positive or negative effect, depending on the sign of the median.

Probability of Direction



PD of cookies

```
plot(density(posterior_cookies))
```



Where is the main mass of the posterior distribution located?
Above zero or below zero?

A simple way of computing PD

```
mean(posterior_cookies > 0)
[1] 0.9865
```

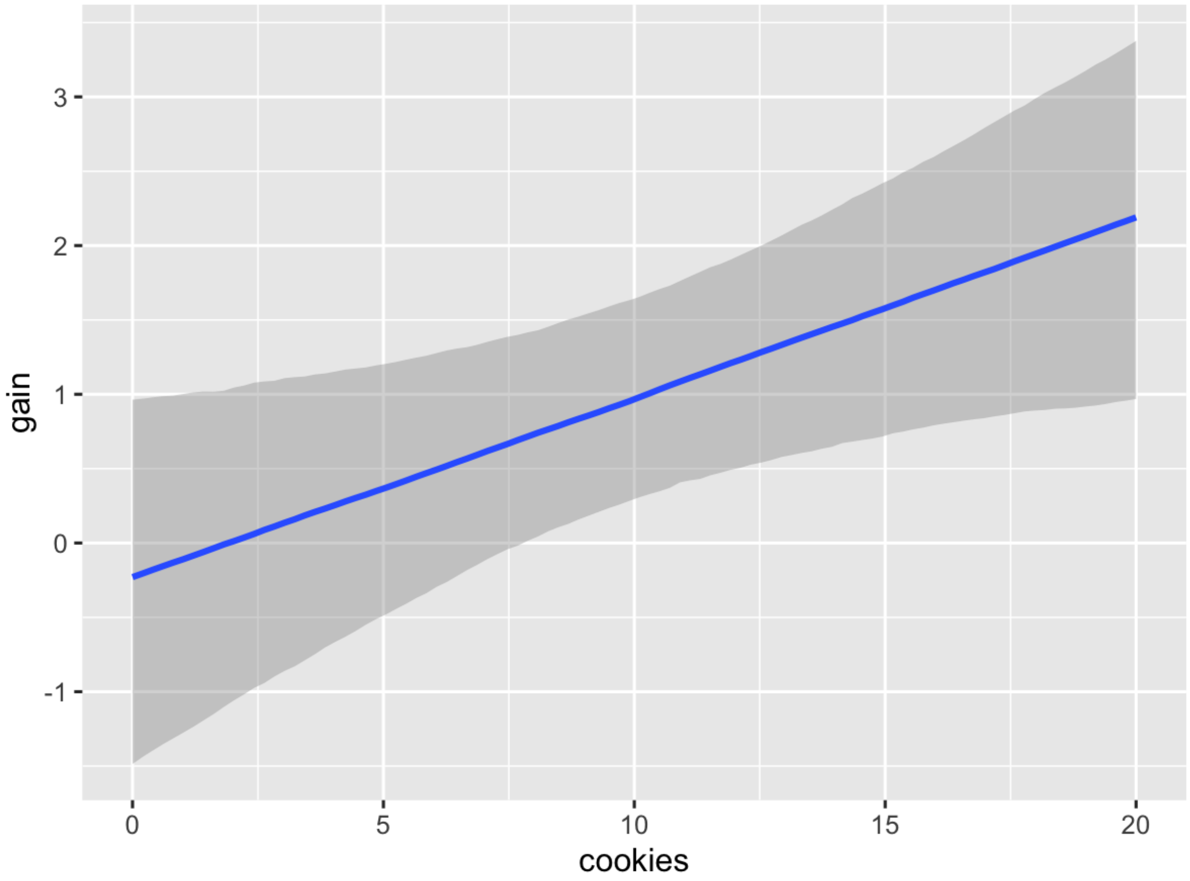
PD = 98.7%

This means that the probability of a positive effect of cookies on weight gain in this model is 98.7%.

A warning: PD can range from 50% to 100%, but it can concentrate around 70% even if the null hypothesis of no effect is true. From experience: when PD is less than 85%, it is not informative and shouldn't be taken seriously.

Visualization of effects

```
conditional_effects(m_brm2, effects = "cookies")
```



Model selection

- OLS and maximul likelihood regression: AIC

- Bayesian regression:
  - WAIC: Watanabe-Akaike Information Criterion
  - LOOIC: Leave-One-Out crossvalidation Information Criterion

Adding a second predictor

```
m_brm3 <- update(m_brm1, formula = . ~ . + sports, ,
newdata = df_cookies)
```

Model selection with WAIC

```
m_brm1 <- add_criterion(m_brm1, criterion = "waic")
m_brm3 <- add_criterion(m_brm3, criterion = "waic")
loo_compare(m_brm1, m_brm3, criterion="waic")
       elpd_diff se_diff
m_brm3  0.0       0.0
m_brm1 -3.0       1.9
```

The model on top is the better one!

How to interpret ELPD diff?

- The elpd quantity stands for expected log predicted density, a measure of accuracy.
- The negative difference between the new and the old model means that the model with sports is slightly better.
- However, the difference is too small given the error. As a very informal rule of thumb, the difference should be at least 2 times as large as standard error, for one model to be preferred over the other. Some people even recommend 5 times. If this is not the case, it make more sense to prefer a more parsimonious model.

Using LOOIC

```
m_brm1 <- add_criterion(m_brm1, criterion = "loo")
m_brm3 <- add_criterion(m_brm3, criterion = "loo")
loo_compare(m_brm1, m_brm3, criterion="loo")
      elpd_diff se_diff
m_brm3   0.0       0.0
m_brm1  -2.8       1.9
```

Note: if there are problems (Pareto k > 0.7), it's best to use more draws. It is also often recommended to run k-fold cross-validation (criterion = "kfold"). This usually takes a lot of time.

Model comparison and selection

Frequentist statistics	Bayesian statistics
Likelihood ratio test, AIC; the models must be nested	LOOIC, WAIC et al.; the models do not have to be nested (but should be fitted on the same dataset)

Goodness of fit

```
bayes_R2(m_brm1)
 Estimate Est.Error       Q2.5      Q97.5
R2 0.4669807  0.181622 0.02227981 0.6722972


bayes_R2(m_brm3)
Estimate Est.Error      Q2.5      Q97.5
R2 0.7286472 0.1252376 0.3601212 0.8287746
```

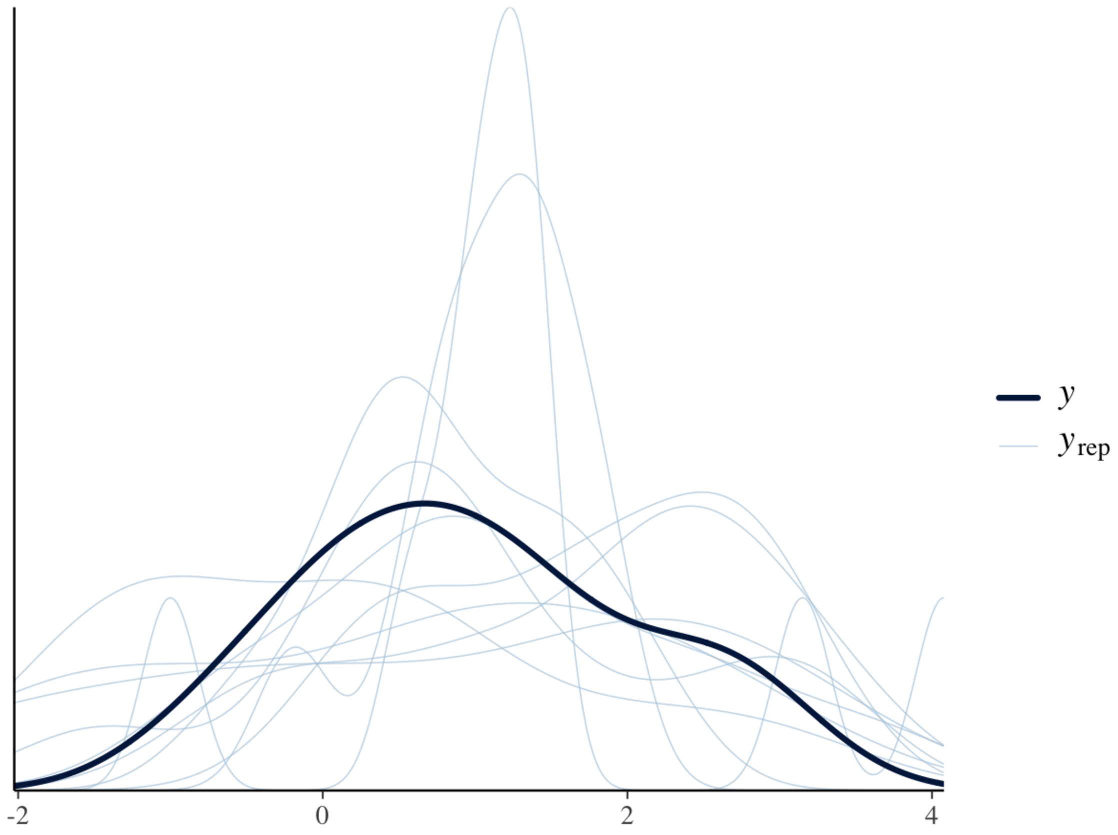
Fitted values

```
fitted(m_brm3)
       Estimate Est.Error       Q2.5     Q97.5
 [1,] -0.7225321 0.4978829 -1.70606642 0.2472253
 [2,]  0.7488068 0.4395060 -0.16041896 1.6384363
 [3,] -0.2249250 0.4024108 -1.04633011 0.5592176
 [4,]  1.1468925 0.3453788  0.44075886 1.8531487
 [5,]  1.3459353 0.3164064  0.69724085 2.0018856
 [6,]  0.2726821 0.3873068 -0.52045475 1.0634715
 [7,]  1.7440210 0.3119263  1.09594266 2.3796009
 [8,]  2.0425852 0.3553551  1.34581634 2.7791150
 [9,]  1.0688534 0.5337544 -0.02844727 2.1522711
[10,]  2.4406709 0.4542799  1.56218395 3.3462904
```

Posterior predictive check

- If my fitted model generated new data, would those data look like the observed data?
- The model pulls parameter estimates (like coefficients and standard deviations) from the MCMC posterior draws.
- It uses these parameters to generate multiple "fake" datasets, accounting for both parameter and residual uncertainty.
- See `?posterior_epred.brmsfit`

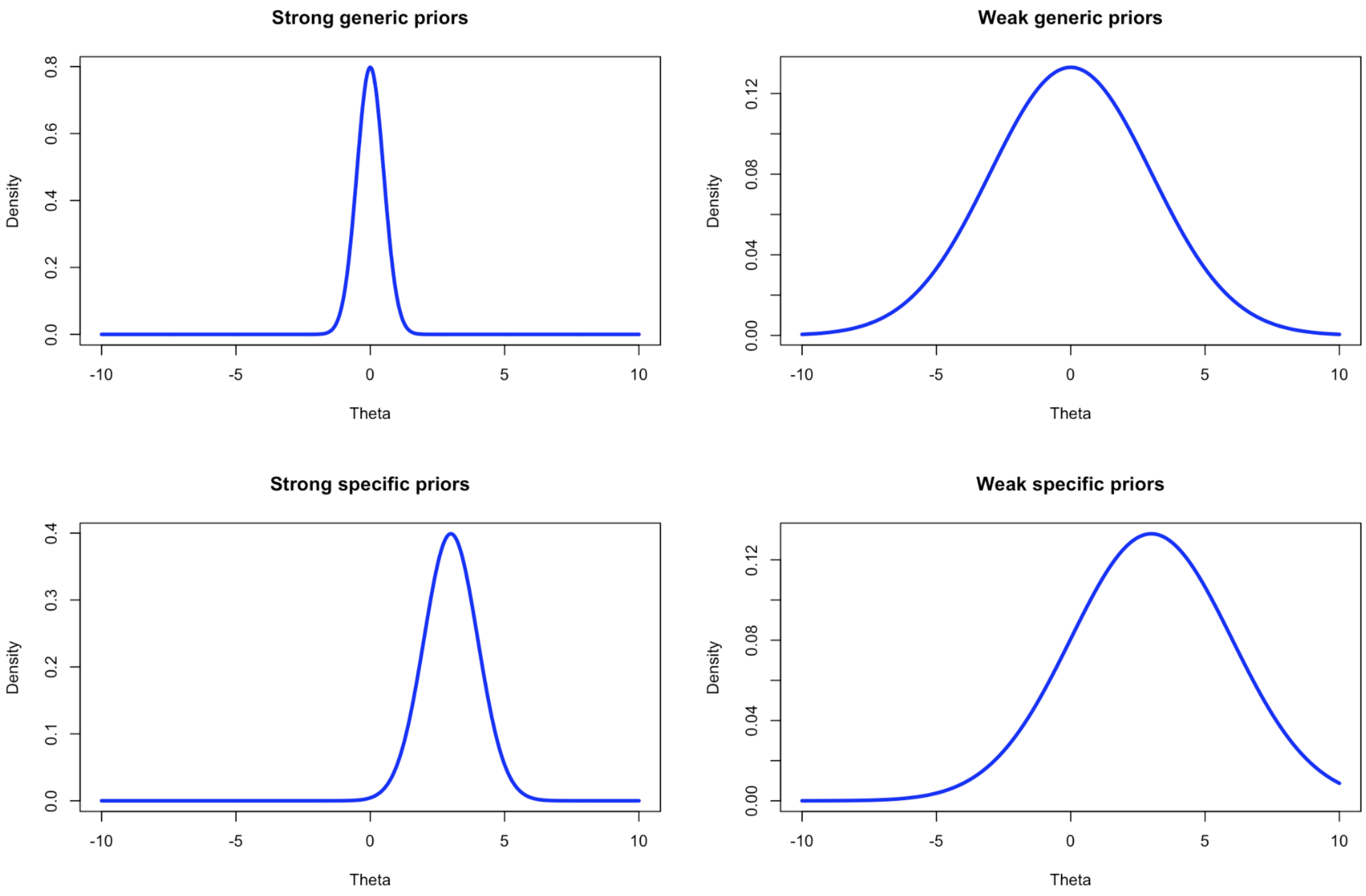
```
pp_check(m_brm1)
```



Model diagnostics

Frequentist statistics	Bayesian statistics
a checklist of possible violations	understanding if new data generated fitted model would look similar to the original data; if not, change the model

Types of priors



The default priors

```
get_prior(gain ~ cookies + sports, data = df_cookies)
```

prior	class	coef	group	resp	dpar	nlpar	lb	ub	source
(flat)	b								default
(flat)	b	cookies							(vectorized)
(flat)	b	sportsYes							(vectorized)
student_t(3, 0.9, 2.5)	Intercept								default
student_t(3, 0, 2.5)	sigma						0		default

Student's t priors in brms

- Student's t priors are used automatically for the intercept in brms.
- There are technical reasons for doing so. Be careful and don't change them unless necessary.
- The numbers, e.g., student_t(3, 0, 2.5) mean the following:
  - 3 degrees of freedom determining the shape.
  - 0 non-centrality parameter, or ncp. Determined automatically: median of the response variable median(y).

    ```
    median(gain)
    [1] 0.9
    ```

  - 2.5 is the default scaling parameter for mad(y) less than 2.5. If it is greater than 2.5, then the actual mad(y) is taken.

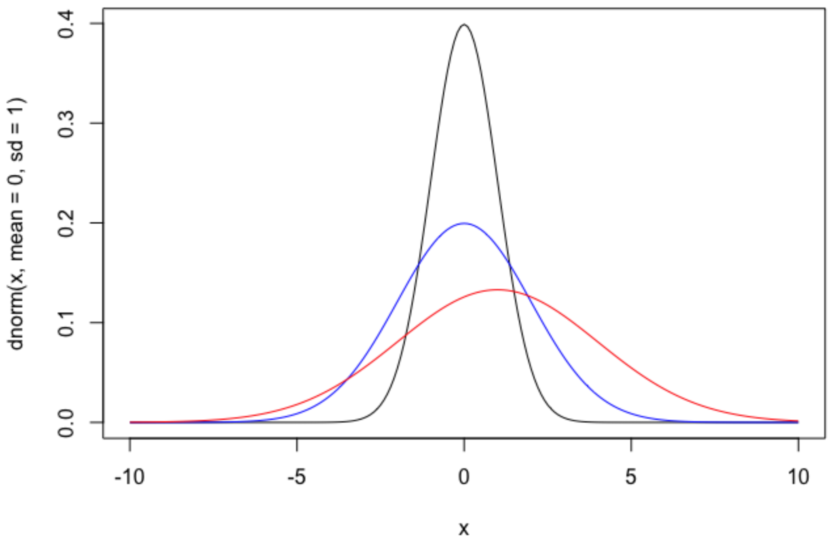
Priors of a fitted model

```
prior_summary(m_brm3)
```

```
                   prior     class      coef group resp dpar nlpar lb ub tag       source
                  (flat)         b                                                default
                  (flat)         b   cookies                                  (vectorized)
                  (flat)         b sportsYes                                  (vectorized)
 student_t(3, 0.9, 2.5) Intercept                                                 default
   student_t(3, 0, 2.5)     sigma                                  0              default
```

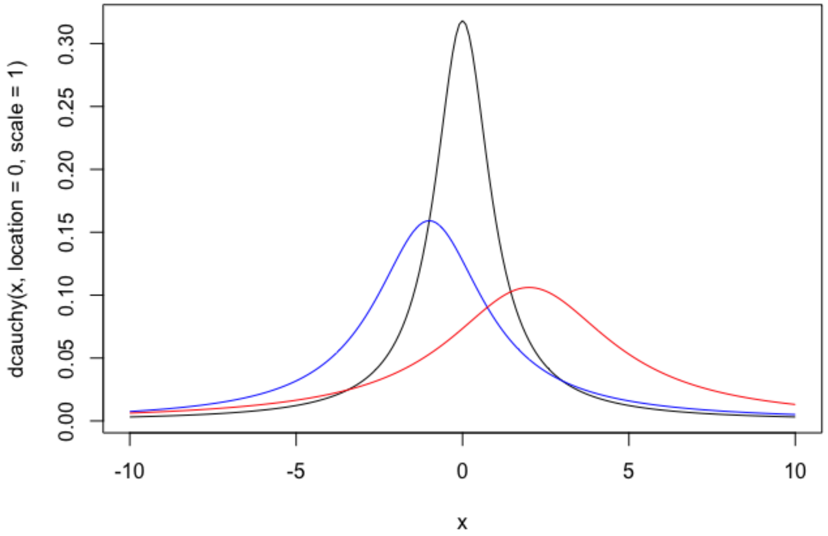
Choosing priors: Normal

```
x <- seq(-10, 10, by = 0.1)
plot(x, dnorm(x, mean = 0, sd = 1),type="l")
lines(x, dnorm(x, mean = 0, sd = 2), col = "blue")
lines(x, dnorm(x, mean = 1, sd = 3), col = "red")
```



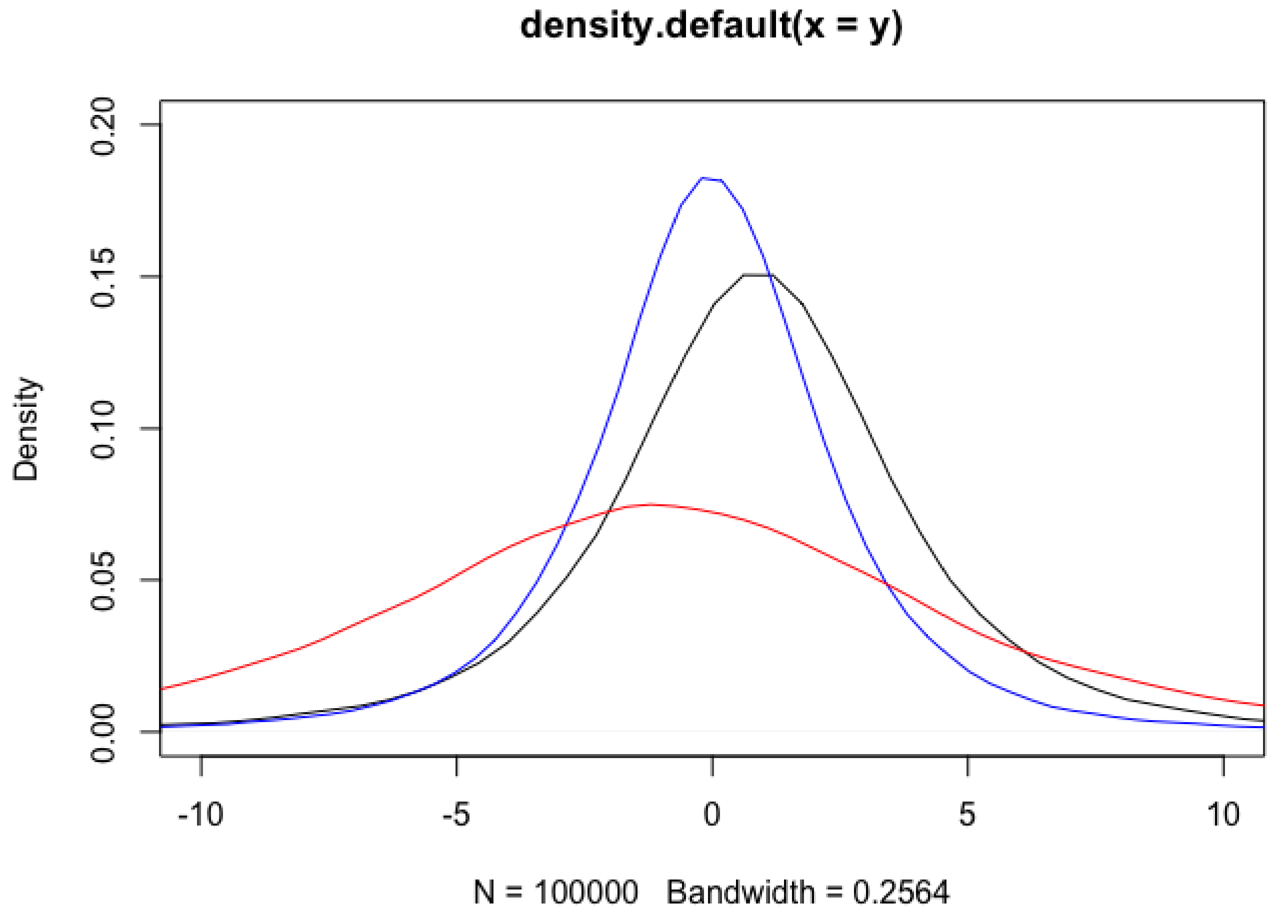
Choosing priors: Cauchy

```
plot(x, dcauchy(x, location = 0, scale = 1), type = "l")
lines(x, dcauchy(x, location = -1, scale = 2), col = "blue")
lines(x, dcauchy(x, location = 2, scale = 3), col = "red")
```



Choosing priors: Student's t

```
y <- rt(100000, df = 3)*2.5 + 0.9
#the values from our model
plot(density(y), xlim = c(-10, 10), ylim = c(0, 0.2))
y <- rt(100000, df = 3)*2 + 0
lines(density(y), col = "blue")
y <- rt(100000, df = 4)*5 - 1
lines(density(y), col = "red")
```


density.default(x = y)
Density
0.00
0.05
0.10
0.15
0.20
-10
-5
0
5
10
N = 100000   Bandwidth = 0.2564

Considerations when choosing priors

- How thick should the tails be? Thick tails allow for more freedom (e.g., Cauchy is more flexible than Normal).
- What about the mean/location/non-centrality parameter? Zero or do you have expectations about the most likely values?
- Note that poorly chosen priors (e.g., too strict, in the wrong direction) can have a bad impact on the chains.

Model with default priors: posteriors

```
summary(m_brm3)
…
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     0.45      0.53    -0.66     1.54 1.00     2715     1987
cookies       0.10      0.04     0.03     0.17 1.00     3120     2381
sportsYes    -1.17      0.51    -2.16    -0.13 1.00     2792     2105
```

Defining priors for all betas

Let's use Normal priors with the mean 0 and standard deviation of 1 for **all** beta coefficients. These are generic priors, but they are quite strong.

```
my_priors1 <- prior(normal(0, 1), class = "b")
m_brm4 <- brm(gain ~ cookies + sports, data =
df_cookies, prior = my_priors1)
summary(m_brm4)
```

Coeffients of the fitted model

What has changed?

```
Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     0.32      0.52    -0.76     1.30 1.00     2970     1959
cookies       0.10      0.04     0.03     0.18 1.00     3009     2290
sportsYes    -0.96      0.44    -1.77    -0.02 1.00     3133     2269
```

Informative strong priors

Normal priors with the mean 3 and SD of 1 are informative and strong.

```
my_priors2 <- prior(normal(3, 1), class = "b")
m_brm5 <- brm(gain ~ cookies + sports, data = df_cookies, prior =
my_priors2)
summary(m_brm5)
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-0.50	1.01	-2.98	0.93	1.00	1288	878
cookies	0.14	0.07	0.03	0.30	1.01	1799	1107
sportsYes	0.24	0.95	-1.16	2.46	1.00	891	1560

Informative but weaker priors

Cauchy's priors with the location parameter 3 and the scaling parameter 2.

```
my_priors3 <- prior(cauchy(3, 2), class = "b")
m_brm6 <- brm(gain ~ cookies + sports, data =
df_cookies, prior = my_priors3)
summary(m_brm6)
```

```
 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept      0.37      0.54    -0.81     1.41 1.00     2499     1852
cookies        0.10      0.04     0.03     0.18 1.00     2733     2103
sportsYes     -1.05      0.50    -2.00     0.07 1.00     2707     1684
```

Different priors for specific predictors

```
my_priors4 <- c(prior(normal(0, 1), class = "b", coef =
"cookies"), prior(cauchy(2, 1), class = "b", coef =
"sportsYes"))

m_brm7 <- brm(gain ~ cookies + sports, data =
df_cookies, prior = my_priors4)

summary(m_brm7)
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.37	0.54	-0.77	1.39	1.00	2878	2003
cookies	0.10	0.04	0.03	0.18	1.00	2900	1981
sportsYes	-1.03	0.51	-1.99	0.04	1.00	3060	2125

Prior sensitivity analysis

- The tests we have performed are called prior sensitivity analysis.
- We see that the posteriors can be strongly influenced by the choice of priors.
- This can happen when the data is small.
- It is always recommended to perform and report a prior sensitivity analysis, but when the sample size is small, it is absolutely crucial.

Exercise

- Try to define informative priors for Sports and Cookies, such that eating more cookies helps you lose weight, and doing Sports leads to gaining weight!

Outline

- Basic principles: Q&A

- Linear regression:
  - ordinary least squares model with lm()
  - Bayesian model with brm()

- Simulating data for mixed models

- Linear mixed regression:
  - (restricted) maximum likelihood model with lmer()
  - Bayesian model with brm()

Data simulation

- See the R file.

Outline

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- Linear mixed regression:
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  - Bayesian model with brm()

Fitting a LMM with lme4

```
library(lme4) #you may need to install it first
mm_lmer <- lmer(Nwords ~ (1|Student_ID) + (1|Topic_ID) + Vocab_Score + University, data = df)
summary(mm_lmer)
Linear mixed model fit by REML ['lmerMod']
Formula:
Nwords ~ (1 | Student_ID) + (1 | Topic_ID) + Vocab_Score + University
Data: df

REML criterion at convergence: 7646.5
```

Outline

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Fitting a LMM with brms

```
mm_brm <- brm(Nwords ~ (1|Student_ID) + (1|Topic_ID) + Vocab_Score + University, data = df, cores = 4)
summary(mm_brm)
```

```
 Family: gaussian
  Links: mu = identity; sigma = identity
Formula: Nwords ~ (1 | Student_ID) + (1 | Topic_ID) + Vocab_Score + University
   Data: df (Number of observations: 1000)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000
```

Default priors for mixed models

```
get_prior(Nwords ~ (1|Student_ID) + (1|Topic_ID) + Vocab_Score +
University, data = df)
```

```
                prior     class        coef      group resp dpar nlpar lb ub       source
               (flat)         b                                                    default
               (flat)         b University1                                   (vectorized)
               (flat)         b Vocab_Score                                   (vectorized)
 student_t(3, 137, 16.3) Intercept                                                 default
   student_t(3, 0, 16.3)       sd                                         0        default
   student_t(3, 0, 16.3)       sd             Student_ID                  0   (vectorized)
   student_t(3, 0, 16.3)       sd   Intercept Student_ID                  0   (vectorized)
   student_t(3, 0, 16.3)       sd               Topic_ID                  0   (vectorized)
   student_t(3, 0, 16.3)       sd   Intercept   Topic_ID                  0   (vectorized)
   student_t(3, 0, 16.3)    sigma                                         0        default
```

What is new?

Defining priors for LLM: an example

```
my_priors_mm <- c(prior(normal(0, 2), class = "b", coef
= "University1"), prior(normal(0, 1), class = "sd",
group = "Student_ID"))
```

Useful when fitting maximal models with **both** random intercepts and slopes:

```
my_priors <- c(… , prior(lkj(2), class = "cor"))
```

A tip

```
Warning message:
There were 1 divergent transitions after warmup. Increasing adapt_delta above 0.8 may
 help. See http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

Try `brm(… , control = list(adapt_delta = 0.9))` or higher values.

Exercise

- Using the tools discussed earlier today, interpet and assess the Bayesian mixed model:
  - Visualize the effects and interpret them
  - Estimate goodness of fit
  - Perform model diagnostics
  - Compare models with and without University using WAIC. Which model is better?
  - Use strong informative priors to "flip" the effect of University